F78PA Probability & Statistics A

Specimen Midterm test

1. A factory produces light bulbs from two machines: Machine A and Machine B. Machine A produces 60% of the total output, and Machine B produces the remaining 40%. The probability that a bulb from Machine A is defective is 0.02, while the probability that a bulb from Machine B is defective is 0.05.

   Let the random variable X represent whether a randomly selected bulb is defective ($X = 1$) or not defective ($X = 0$).

   (a) Define a suitable outcome space Ω that captures both the machine used and whether the bulb is defective or not. (1 mark)

   (b) Let $\mathcal{F}$ be a set of subsets of Ω. State the properties $\mathcal{F}$ must satisfy to be a valid event space. (2 marks)

   (c) Calculate the probability that a randomly selected bulb is defective, i.e $X = 1$. (2 marks)

   (d) Given that a bulb is defective, calculate the probability that it was produced by Machine B. (2 marks)

   (e) Are the events that the bulb is defective and the bulb is produced by Machine A independent? Justify your answer mathematically. (3 marks)

   [Total 10 marks]

2. A company manufactures metal rods used in construction. The length of a randomly selected rod, measured in metres, is modelled by a continuous random variable X. Quality control analysis suggests that the distribution of rod lengths follows the probability density function:

$$f_X(x) = \begin{cases} kx(2-x), & \text{for } 0 \leq x \leq 2 \\ 0, & \text{otherwise .} \end{cases}$$

   (a) State the definition of a probability density function for a continuous random variable X. (1 mark)

   (b) Determine the value of the constant k that makes f_X a valid probability density function. (2 mark)

   (c) Find and clearly define the cumulative distribution function, F_X, of X. (3 marks)

   (d) Calculate the probability that a randomly selected rod has length between 0.5 and 1.5 meters. (1 marks)

   (e) The mode of a random variable is point x^* which maximises the probability density function. Find the mode of X. (3 marks)

   [Total 10 marks]